

Instructions and submission rules

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* Indicates required question

Technical plan

In this section, describe how you plan to build the project technically.

Your answer should include the proposed stack, architecture direction, and main implementation choices such as:

- backend,
- frontend,
- AI/ML tools if any,
- database/storage,
- Docker/containerization,
- and cloud/deployment approach.

The goal is not to have every detail finalized, but the plan should already be technically reasonable.



Proposed technical stack *

List backend, frontend, AI/ML tools if any, database, deployment/container tools, and cloud services.

Backend: Java (Spring Boot) for building robust, decoupled microservices.

Frontend: React.js with TypeScript to build dynamic and responsive staff and reporting dashboards.

Database/Storage: PostgreSQL as the primary relational database (ideal for ACID transactions on inventory and orders), Redis for caching and session management, and an S3-compatible object store (e.g., MinIO or AWS S3) for document/invoice storage.

Deployment/Containers: Docker, Docker Compose, and GitHub Actions for automated CI/CD pipelines.

Cloud Services/Observability: AWS (ECS/EC2) for hosting, with Prometheus and Grafana integrated for system-wide monitoring and logging.

Will the system include a backend with APIs? *

- ☒ Yes
- ☐ No
- ☐ Not Applicable

Will the project include a database or persistent storage? *

- ☒ Yes
- ☐ No
- ☐ Not Applicable



Will the project be Dockerized? *

- ☒ Yes
- ☐ No
- ☐ Planned but not finalized

Planned cloud/deployment approach *

Describe how the project will be deployed or demonstrated in the cloud.

The system will be deployed using a containerized, zero lock-in microservices architecture. We will use Docker to containerize all 6 core modules and deploy them via a managed container service. The deployment lifecycle will be fully automated using CI/CD pipelines to enforce testing and streamline releases. To meet the core cloud requirements, we will implement centralized observability using Prometheus for metric collection and Grafana for monitoring system health. The architecture assumes horizontal scalability; services will remain stateless, allowing us to spin up additional instances of high-traffic modules (like Order Processing) under load.

For AI-related projects: dataset / input source / simulation source

State the dataset, source, size, and availability. If not AI, describe the test data or input scenarios.

As this is not an AI project, we will utilize simulated business operations data. > We will generate a comprehensive mock dataset to rigorously test our microservice boundaries and scaling assumptions. This will include a catalog of at least 1,000 unique inventory SKUs, 50 configured warehouse storage zones, and a set of automated load-testing scripts. These scripts will simulate concurrent user activity (e.g., generating 100+ simultaneous orders) to validate database locking mechanisms, ensure accurate inventory deductions, and test the resilience of our API contracts under stress.

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